

RMSE, MAE, MSE, R^2

(where we predict a continuous value such as house price, stock price, sales amount, temperature, etc.).

Let's use a very simple example.

Actual Value	Predicted Value
100	90
200	220
300	280
400	410

Step 1: Calculate Errors

Error = Actual – Predicted

Actual	Predicted	Error
100	90	10
200	220	-20
300	280	20
400	410	-10

Notice some errors are positive and some are negative.

If we simply average them:

$$(10 - 20 + 20 - 10) / 4 = 0$$

Average error becomes zero, which is misleading because the model is clearly making mistakes.

That's why MAE, MSE, RMSE were invented.

1. MAE (Mean Absolute Error)

MAE measures the average absolute difference between actual and predicted values.

$$MAE = \frac{\sum |Actual - Predicted|}{n}$$

Calculate

Absolute Errors:

Error	Absolute Error
10	10
-20	20
20	20
-10	10

$$MAE = \frac{(10 + 20 + 20 + 10)}{4}$$

$$MAE = \frac{60}{4}$$

$$MAE = 15$$

Interpretation

On average, predictions are off by **15 units**.

If predicting house prices in lakhs:

"My predictions are wrong by ₹15 lakh on average."

Advantages

- ✓ Easy to understand
- ✓ Same unit as target variable
- ✓ Less affected by outliers

Disadvantages

- ✗ Doesn't heavily punish large mistakes

2. MSE (Mean Squared Error)

Instead of taking absolute value, square the errors.

$$MSE = \frac{\sum (Error)^2}{n}$$

Calculate

Err or	Squared Error
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10	100
----	-----

-20	400
-----	-----

20	400
----	-----

-10	100
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$$MSE = \frac{(100 + 400 + 400 + 100)}{4}$$

$$MSE = 250$$

Why Square?

Suppose one prediction is off by 100.

$$100^2 = 10000$$

Large errors become very large.

Thus MSE punishes large mistakes more heavily.

Advantages

✓ Excellent for optimization

✓ Penalizes large errors strongly

Disadvantages

✗ Unit becomes squared

For house price:

$$(\text{Rupees})^2$$

which is difficult to interpret.

3. RMSE (Root Mean Squared Error)

RMSE solves the unit problem of MSE.

Take square root of MSE.

$$RMSE = \sqrt{MSE}$$

For our example:

$$RMSE = \sqrt{250}$$

$$RMSE \approx 15.8$$

Interpretation

Average prediction error is approximately **15.8 units**.

Since square root is taken, units return to original scale.

Advantages

- ✓ Same unit as target
- ✓ Penalizes large errors
- ✓ Most commonly used in industry

Disadvantages

- ✗ Sensitive to outliers

MAE vs RMSE Example

Imagine two models:

Model A Errors

10, 10, 10, 10

Model B Errors

1, 1, 1, 37

MAE:

Model	MAE
A	10
B	10

Same MAE.

But RMSE:

Model	RMSE
A	10
B	18.5

RMSE reveals that Model B made one huge mistake.

That's why RMSE is often preferred when large errors are costly.

4. R^2 (R-Squared)

MAE, MSE, RMSE tell:

"How much error exists?"

R^2 tells:

"How much variance in the data is explained by the model?"

Range:

$-\infty$ to 1

Typically:

R^2 Value	Meaning
1	Perfect prediction
0.9	Excellent
0.8	Very good
0.6	Good
0	No predictive power
<0	Worse than predicting average

Intuitive Example

Suppose students' marks vary from:

40, 50, 60, 70, 80

There is variance in the data.

If your model can explain 90% of this variation:

$$R^2 = 0.9$$

Meaning:

Model explains 90% of the behavior of the target variable.

Only 10% remains unexplained.

Formula

$$R^2 = 1 - \frac{SS_{\{Residual\}}}{SS_{\{Total\}}}$$

Where:

- $SS_{Residual}$ = Unexplained variation (errors)
- SS_{Total} = Total variation in data

Think of it as:

$$R^2 = 1 - \frac{\{Error\}}{\{Total\ Variation\}}$$

Business Interpretation

Suppose you build a model predicting AWS monthly cloud spend.

Metric	Result
MAE	₹5,000
RMSE	₹8,000
R^2	0.92

Meaning:

- Average prediction error = ₹5,000
- Large mistakes exist occasionally ($RMSE > MAE$)
- Model explains 92% of spend variation

This would generally be considered a strong model.

Quick Comparison

Metric	Measures	Lower/ Higher Better?	Outlier Sensitive?
MAE	Average error	Lower	No
MSE	Squared error	Lower	Yes
RMSE	Root of MSE	Lower	Yes
R ²	Explained variance	Higher	Indirectly

Easy Interview Answer

MAE

- Average absolute prediction error.
- Easy to interpret.

MSE

- Average squared error.
- Punishes large mistakes heavily.

RMSE

- Square root of MSE.
- Same unit as target variable.
- Most widely used regression metric.

R²

- Measures how much variance the model explains.
- Closer to 1 means a better fit.

A common practical approach is:

Use **RMSE + R² together**. RMSE tells you the magnitude of prediction errors, while R² tells you how well the model explains the data.